

Open and Sort Mail SOP

One episode opens three sealed envelopes with an opener, sorts what comes out of each one into two trays, and flattens one small box for recycling. The three loaded envelopes are standing in the mail holder and the box is standing at the box spot when recording starts. Each envelope is taken to the work spot, slit along its flap edge with the opener, emptied, and cleared to the recycling tray. The card goes to the card tray, the sheet is unfolded flat and goes to the sheet tray. When all three are done the box top flaps are folded out, the box is pressed flat, and the flat box is laid in the recycling tray.

The order never changes. The envelopes are taken from the front of the mail holder, one at a time, and one envelope is finished before the next one is touched: taken, slit, emptied, card away, sheet unfolded, sheet away. The box is not touched until all three envelopes are done.

Every envelope is opened with the opener. Nothing is ever torn, ripped, or pulled open. Every envelope is slit lying flat on the desk, never held up in the air, and the opener goes back to the opener rest before either gripper touches anything else.

The right gripper does the primary work: it takes each envelope to the work spot (in Config R1 and R2, below), works the opener, slits, draws the contents out, clears the empty envelope, unfolds the sheet, and folds and presses the box. The left gripper supports: it takes each envelope to the work spot in Config M, pins the envelope, the sheet, and the box while the right gripper works, folds the left top flap of the box, and carries the card and the sheet to the trays on the left. Both grippers together carry the flat box. Neither gripper takes the other's jobs.

The desk is set up in one of three ways. Only the mail holder with its three envelopes moves; the opener rest, the work spot, the box spot, the sheet tray, and the card tray are in the same place in all three. The recycling tray moves only in Config M, where the mail holder takes its place.

- **Config M:** the mail holder stands at the front center, in front of the work spot, and the recycling tray is at the front right.
- **Config R1:** the mail holder stands at the front right.
- **Config R2:** the mail holder stands at the back right.

Where a step depends on the setup it says so on an **IF** line — look at the desk and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the mail holder starts at the front center (**Config M**), the front right (**Config R1**), or the back right (**Config R2**). One config per episode, chosen before recording and never changed mid-episode. The left side is not used: both left corners are the left gripper's

trays, and the sheet tray, which takes a fully open sheet, leaves no left corner free for the mail holder.

- **Same-side rule:** the gripper on the mail holder's side takes each envelope to the work spot — the **left gripper** in Config M, the **right gripper** in Config R1 and R2. No arm reaches across the desk for an envelope.
- **Fixed roles:** everything else is the same in all three configs — the right gripper works the opener, slits, draws the contents out, clears the empty envelope, unfolds the sheet, and folds and presses the box; the left gripper pins, folds the left top flap, and carries the card and the sheet to the trays on the left; both grippers carry the flat box.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the mail holder in the start zone for this episode's config, the opener rest right of the work spot, the work spot at the center, the box spot at the back center, the sheet tray at the back left, the card tray at the front left, and the recycling tray at the front center (front right in Config M).
3. The overhead view shows the whole work spot, so the flap edge of an envelope, the slit running along it, and every panel of an unfolded sheet can be read.
4. Both arms are at home with grippers open.
5. The desk is clear of anything but the mail holder and its three envelopes, the opener on its rest, the box, and the three trays.
6. The right gripper reaches (Config R1 and R2) the front envelope in the mail holder, the opener rest, the whole work spot including the left end of an envelope lying there, the box spot, and the recycling tray, without the arm leaning out or reaching a joint limit.
7. The left gripper reaches (Config M) the front envelope in the mail holder, the whole work spot, the sheet tray, the card tray, the box spot, and the recycling tray, without the arm leaning out or reaching a joint limit.

Materials checklist

1. Three sealed **envelopes**, all the same size, standing upright in the **mail holder** in the start zone for this episode's config, leaning back, **flap side toward the front edge**, in a fixed order. Envelope 1 is the one nearest the front edge, then envelope 2, then envelope 3.

- **Config M:** front center, in front of the work spot, where the recycling tray normally sits
 - **Config R1:** front right, in front of the opener rest
 - **Config R2:** back right
2. Each envelope is plain, uncrushed, dry, and unslit, with its **flap** sealed along its long top edge and lying flat.
 3. Each envelope is sealed so that the **corner gap** at the right end of the flap edge stays open, wide enough for the opener tip to enter without being forced.
 4. Each envelope holds one **sheet** and one **card**, and nothing else. The same two items go into every envelope.
 5. The **sheet** is one plain page folded in three by a roll fold, so it lies as three stacked panels: a **top panel** on top, a **tucked panel** under it hinged on the opposite side, and a **base panel** under both. It is uncreased apart from its two fold edges, unmarked, and it lies flat when opened out rather than springing back.
 6. The **card** is one stiff flat card, clearly smaller than the folded sheet, undamaged and unbent. It is loaded on top of the folded sheet, so the two come out of the envelope as one bundle with the card on top.
 7. The sheet is loaded with the **free edge of its top panel toward the flap edge**, so that edge leads coming out of the envelope.
 8. One **opener** lies on the **opener rest** right of the work spot, along the desk parallel to the front edge, **handle at the right end and blade pointing left**. It is a blunt letter opener with a smooth **tip** that parts a seal under one steady draw. Its handle is thick enough for the gripper to hold without the opener turning in the fingers.
 9. One small **box** stands at the **box spot** at the back center: empty, untaped, uncrushed, square, with all four **bottom flaps** folded outward flat against the desk so the box stands on them, and all four **top flaps** standing up. All four **corner creases** are free enough to fold flat under a steady draw.
 10. The **work spot** at the center of the desk is bare and wide enough that the sheet lies fully open inside it with clear desk all round.
 11. The **sheet tray** at the back left and the **card tray** at the front left are empty, square to the desk, and each is large enough to hold three of its item lying flat with clear tray all round. The sheet tray takes an unfolded sheet without any part of it overhanging.
 12. The **recycling tray** at the front center (front right in Config M) is empty, square to the desk, and large enough to hold the three slit envelopes and the flat box lying flat with clear tray all round.
 13. The mail holder, the opener rest, and all three trays sit firmly enough that a gripper coming down on them does not shift them.

Workspace layout

- **Mail holder:** the start zone — front center (**Config M**), front right (**Config R1**), or back right (**Config R2**). Holds the three loaded envelopes upright and in order. The left gripper's zone in Config M, the right gripper's zone in Config R1 and R2.
- **Opener rest:** right of the work spot. Holds the opener and nothing else. The right gripper's zone.
- **Work spot:** center of the desk. Every slit, every draw, and every unfold happens here. Both grippers reach it, one to pin and one to work.
- **Box spot:** back center, beyond the work spot. Holds the box all episode until it is flattened. Both grippers reach it.
- **Sheet tray:** back left. Receives the three unfolded sheets and nothing else. The left gripper's zone.
- **Card tray:** front left. Receives the three cards and nothing else. The left gripper's zone.
- **Recycling tray:** front center, or front right in Config M, where the mail holder takes the front center. Receives the three slit empty envelopes, then the flat box on top of them. Both grippers reach it.

Everything the right gripper picks up is staged on the right or at the center. The card and the sheet, the only things the left gripper carries to a tray, move along the left. Nothing is carried over the box.

Vocabulary

- **Arm assignment:** the right gripper takes each envelope to the work spot in Config R1 and R2, works the opener, slits, draws the contents out, clears the empty envelope, unfolds the sheet, and folds and presses the box. The left gripper takes each envelope to the work spot in Config M, pins, folds the left top flap of the box, and carries the card and the sheet to their trays. Both grippers together carry the flat box. Neither gripper takes the other's jobs.
- **Start zone:** where the mail holder stands at the start of the episode — front center (**Config M**), front right (**Config R1**), or back right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Flap:** the sealed leaf along the long top edge of the envelope. The **flap edge** is the line it is sealed along, and the **corner gap** is the short unsealed opening at the right end of that line.
- **Slit:** the seal parted along the whole flap edge by one draw of the opener under the flap. Nothing is cut off the envelope and no paper is torn.
- **Cut flap:** the flap once the seal is parted. It is free along its whole length and still joined to the envelope at its hinge.
- **Mouth:** the opening along the flap edge once the cut flap is laid back.
- **Contents:** one sheet and one card. They leave the envelope as one **bundle** with the card on top.

- **Sheet panels:** the folded sheet has three. The **top panel** is the layer facing up, hinged at one fold edge with its **free edge** on the other side. The **tucked panel** lies under it, hinged at the opposite fold edge with its own free edge exposed. The **base panel** lies flat on the desk between the two fold edges.
- **Unfolded sheet:** all three panels lying flat in one line on the desk, none still folded over another, with no corner curled up.
- **Opener parts:** the **handle** the gripper holds, the **blade**, and the **tip** at its end.
- **Tool home:** the opener rest, in its staged position, handle at the right end and blade pointing left. The opener returns there at the end of every slit, before either gripper touches anything else.
- **Box parts:** the four **box walls**, the **top rim** where they end, the four **top flaps** above the rim, the four **bottom flaps** folded flat on the desk, and the four **corner creases** where the walls meet.
- **Flat box:** the box pressed down so all four walls lie flat on the desk, one on another, with no wall standing and no corner crease bowed. Its **left fold edge** and **right fold edge** are the two closed sides it makes.
- **Lay down:** lowering a held envelope until its bottom edge rests on the desk, then swinging its top edge away from the front edge until the whole envelope lies flat, and releasing only once it is resting. An envelope is never turned over in the air.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Pin:** a closed gripper pressed straight down to hold something still. A pin never pinches and never travels.
- **Press:** the closed gripper set straight down flat, held still for 2 seconds, and lifted straight off. A press never slides and never rubs.
- **Stroke:** the closed gripper pressed down and drawn along in one continuous run, end to end, without lifting off.
- **Draw:** a gripper closed on something and pulled along in one continuous run to its end point, without releasing and without lifting off partway.
- **Fold:** a gripper pinching a flap or a panel at the middle of its free edge, swinging it over its fold edge until it lies flat, holding for 2 seconds, and releasing.
- **Settled:** the thing stays put after the gripper lifts clear, with nothing springing back.
- **Release point:** the work spot, the opener rest, the sheet tray, the card tray, the recycling tray, or the box spot. A gripper opens nowhere else.
- **Stray content:** any sheet, card, or envelope resting outside the work spot and outside its tray.

Steps

Steps 1 to 6 are one envelope. Run them in full three times, taking the envelope at the front of the mail holder each time. Steps 7 to 9 are the box and run once, after all three envelopes are done. Step 10 ends the episode.

Only the pick in Step 1 depends on the config: the **left gripper** takes the envelope from the front center in Config M, the **right gripper** from the front right in Config R1 or the back right in Config R2. Every other line is the same in all three configs; in Config M the recycling tray stands at the front right and Steps 3 and 9 carry to it there.

Step 1: Take an envelope to the work spot

Goal: one envelope lying flat and square at the work spot, flap side up, flap edge away from the front edge.

Look where the mail holder is before reaching for the envelope.

- **IF the mail holder is at the front center (Config M):** the **left gripper** closes on the **top edge** of the front envelope in the **mail holder** and **lifts** it straight up until it is clear of the holder.
- **IF the mail holder is at the front right (Config R1):** the **right gripper** closes on the **top edge** of the front envelope in the **mail holder** and **lifts** it straight up until it is clear of the holder.
- **IF the mail holder is at the back right (Config R2):** the **right gripper** closes on the **top edge** of the front envelope in the **mail holder** and **lifts** it straight up until it is clear of the holder.

Then, in all three:

- The same gripper carries the envelope upright to the **work spot** and **lays it down:** it lowers the envelope until its bottom edge rests on the desk, then swings its top edge away from the front edge until the envelope lies flat, and releases once it is resting.
- The same gripper lifts clear.

The envelope goes down onto the desk and tips over there. Turning it over in the air needs a wrist twist the arm does not have, and the envelope slides in the fingers instead.

Check: the envelope lies flat and square at the work spot, flap side up, the flap edge away from the front edge and the bottom edge toward the front edge, sealed and unslit, with no part of it off the work spot and no corner folded under. The mail holder still stands square with the remaining envelopes upright in it. If the envelope landed skewed, folded, or flap side down, take its edge, lift it clear, and lay it down again.

Step 2: Slit the envelope with the opener

Goal: the seal parted along the whole flap edge, the envelope otherwise undamaged, and the opener back on its rest.

- The **left gripper**, closed, **pins** the envelope at the middle of its **bottom edge**, clear of the flap edge, and holds it there for the whole step.
- The **right gripper** closes on the **handle** of the **opener** on the opener rest and **lifts** it straight up, blade still pointing left.
- The **right gripper** brings the **tip** into the **corner gap** at the right end of the flap edge, under the flap, and **draws** the opener along the flap edge in one continuous run to the left end of the envelope, keeping the tip under the flap the whole way.
- The **right gripper** lifts the opener straight up off the envelope, carries it level back to **tool home**, lowers it until it lies flat with the handle at the right end and the blade pointing left, and releases.
- The left gripper lifts clear only after the opener is resting.

The draw is one run under the flap. An opener pushed into the envelope face cuts the sheet and the card inside, and a seal worked at in short jabs tears the paper instead of parting.

Check: the seal is parted along the whole flap edge, the **cut flap** is free along its whole length and still joined at its hinge, no paper is torn and nothing is cut off the envelope, the envelope is still square at the work spot, and the opener lies at tool home with the handle at the right end and the blade pointing left. If the draw stopped short, take the opener again, enter the corner gap again, and draw again. If the tip will not enter the corner gap, or the seal will not part after two correct draws, stop, log it as a system issue, and discard the episode.

Step 3: Draw the contents out and clear the envelope

Goal: the bundle lying flat at the work spot and the empty envelope in the recycling tray.

- The **left gripper**, closed, **pins** the envelope at the middle of its **bottom edge** and holds it there while the contents come out.
- The **right gripper** pinches the **cut flap** at the middle of its free edge, lifts it clear, and lays it back away from the front edge so the **mouth** stands open. Release.
- The **right gripper** reaches into the mouth, closes on the **bundle** at its top edge, and **draws** it straight out away from the front edge in one continuous run until it is clear of the envelope.
- The **right gripper** lowers the bundle flat onto the work spot beyond the envelope, card side up, and releases.
- The **left gripper** lifts clear.
- The **right gripper** pinches the empty envelope at its **bottom edge**, lifts it, carries it level to the **recycling tray**, lowers it flat inside the tray, and releases.
- The **right gripper** lifts clear.

The bundle is drawn out. An envelope tipped up or shaken to empty it drops the card on the desk and creases the sheet on the way.

Check: the sheet and the card lie flat at the work spot with the card on top and the sheet's two fold edges running along the desk parallel to the front edge, nothing is left inside the envelope, the envelope lies flat inside the recycling tray with no part overhanging a tray edge, and there is no stray content on the desk. If the bundle came out skewed or the card slid off the sheet, lift each piece and lay it down again at the work spot, card on top.

Step 4: Put the card in the card tray

Goal: the card lying flat in the card tray, clear of the other cards.

- The **left gripper** closes on the **card** at its edge, **lifts** it straight up off the sheet, and carries it level to the **card tray**.
- The **left gripper** lowers the card until it lies flat inside the tray, clear of any card already there, and releases.
- The **left gripper** lifts clear.

Check: the card lies flat inside the card tray, not on top of another card and not overhanging a tray edge, undamaged and unbent, and the folded sheet is still lying flat and square at the work spot. If the card landed on another card or overhanging, lift it and lay it down again clear of the others. If the sheet was dragged out of place when the card came off, take its edge and lay it square again.

Step 5: Unfold the sheet

Goal: all three panels lying flat in one line at the work spot, none still folded over another.

- The **left gripper**, closed, **pins** the **base panel** at its middle and holds it there for the whole step.
- The **right gripper folds** the **top panel** open: it pinches the panel at the middle of its **free edge**, lifts it clear of the paper below, swings it over its fold edge, lays it flat on the desk beyond that edge, holds it for 2 seconds, and releases.
- The **right gripper folds** the **tucked panel** open the same way, over the opposite fold edge, laying it flat on the desk on the other side of the base panel.
- The **right gripper**, closed, **strokes** along each of the two fold edges in turn, end to end.
- The left gripper lifts clear only after the right gripper is off the paper.

Each panel is lifted clear and laid down. A panel dragged open across the desk rucks the paper under it and creases the sheet along the wrong line.

Check: all three panels lie flat in one line, none folded over another, no corner curled up, no new crease and no tear anywhere on the sheet, and the sheet is still inside the work spot. If a panel sprang back, fold it open again and stroke its fold edge again.

Step 6: Put the sheet in the sheet tray

Goal: the unfolded sheet lying flat in the sheet tray, clear of the other sheets.

- The **left gripper** pinches the **unfolded sheet** at the middle of its near edge, lifts it clear of the desk, and carries it level to the **sheet tray**.
- The **left gripper** lowers the sheet until it lies flat inside the tray, clear of any sheet already there, and releases.
- The **left gripper** lifts clear.

Check: the sheet lies flat inside the sheet tray, fully open with no panel folded over, not on top of another sheet, and with no part of it overhanging a tray edge. The work spot is bare. If the sheet folded over on the way or landed overhanging, lift it, lay it flat again in the tray, and open the folded panel out there.

Take the next envelope and run Steps 1 to 6 again. Go on to Step 7 only when all three envelopes have been run and the work spot is bare.

Step 7: Fold the box top flaps out

Goal: all four top flaps folded outward and lying flat out from their walls, with the box still square on the box spot.

- The **left gripper**, closed, **pins** the box at its **left top rim** and holds it there while the right gripper folds.
- The **right gripper folds** the **far top flap** out: it pinches the flap at the middle of its free edge, swings it away from the front edge and down until it lies flat out from its wall, holds it for 2 seconds, and releases.
- The **right gripper folds** the **right top flap** out to the right the same way, then the **near top flap** out toward the front edge the same way.
- The **right gripper**, closed, **pins** the box at its **right top rim**, the left gripper lifts clear, and the **left gripper folds** the **left top flap** out to the left the same way.
- Both grippers lift clear.

The far flap goes first. With the near flap already down, the arm has to travel out over it to reach the far one, which drags the near flap back up and puts the arm where it should never be.

Check: all four top flaps lie flat out from their walls, none standing up, curling, or folded inward into the box, the box still stands square on the box spot with all four walls upright and all four bottom flaps still flat on the desk, and the box is still empty. If a flap sprang back up, fold it again and hold it longer before releasing.

Step 8: Press the box flat

Goal: the box lying flat on the desk with all four walls down and both corner fold edges settled.

- The **left gripper**, closed, **pins** the box at the **near bottom crease**, where the near wall meets the desk, and holds it there while the box comes down.
- The **right gripper** closes on the **top rim** of the **far wall** at its middle and **draws** it toward the front edge in one continuous run, lowering it as it comes so the far wall folds down onto the near wall and the box goes flat on the desk. Release once it is down.
- The **right gripper**, closed, **strokes** along the **right fold edge** from its near end to its far end.
- The **left gripper** lifts off the crease and, closed, **strokes** along the **left fold edge** from its near end to its far end.
- Both grippers lift clear.

The box is drawn flat along its corner creases. A box shoved over or pressed straight down from above buckles a wall instead of folding it and leaves a box that will not lie flat.

Check: the box lies flat on the desk, all four walls down with none standing and none buckled, both fold edges straight and settled, all eight flaps flat and none trapped folded under the box, and nothing under the box. If a wall sprang back up or a fold edge bowed, draw the box flat again and stroke both fold edges again.

Step 9: Put the flat box in the recycling tray

Goal: the flat box lying flat inside the recycling tray, on top of the three envelopes.

- The **left gripper** pinches the flat box at its **left fold edge** and the **right gripper** pinches it at its **right fold edge**, each at about the middle of its length.
- Both grippers lift the flat box level off the desk, carry it level to the **recycling tray**, lower it until it lies flat inside the tray on top of the three envelopes, square to the tray edges, and release together.
- Both grippers lift clear.

Both grippers carry the box and release together. A flat box carried on one gripper hangs from that hold and springs open in the air.

Check: the flat box lies flat inside the recycling tray on top of the three envelopes, square to the tray edges, still flat with no wall sprung back up, with no part of it overhanging a tray edge, and the box spot is bare. If it landed skewed or a wall sprang open, take the same two fold edges, lift it clear, press it flat again on the desk, and lay it down again.

Step 10: Confirm the trays and end the episode

- Confirm the sheet tray holds three sheets lying flat and fully open, the card tray holds three cards lying flat, and the recycling tray holds three slit empty envelopes with the flat box lying flat on top of them.

- Confirm the opener lies at tool home, handle at the right end and blade pointing left, the mail holder is empty, and the work spot, the box spot, and the rest of the desk are bare.
- Return both arms home, clear of the work spot and every tray, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the flat box out of the recycling tray, open it back out square, and stand it at the box spot with all four bottom flaps folded outward flat and all four top flaps standing up. Replace a box whose corner creases are crushed, that will not square up, or that will not stand on its bottom flaps.
2. Lift the three slit envelopes out of the recycling tray and throw them away. Fresh envelopes are used every episode. Confirm the tray is empty.
3. Take the three sheets out of the sheet tray and the three cards out of the card tray. Replace any sheet that is torn, marked, or creased anywhere but on its two fold lines, and any card that is bent, split, or dog eared.
4. Fold each sheet back in three by a roll fold, on its existing fold lines, and press it flat so it lies as three stacked panels rather than springing open.
5. Load one folded sheet into each fresh envelope with the free edge of its top panel toward the flap edge, lay one card on top of the sheet, and seal the flap so that the corner gap at the right end of the flap edge stays open.
6. Stand the three loaded envelopes in the mail holder, flap side toward the front edge, in order, and stand the holder in the start zone for the next episode's config — front center (Config M), front right (Config R1), or back right (Config R2). Confirm the holder stands square.
7. Wipe the opener and check its tip and blade. Replace an opener whose tip is bent, burred, or too blunt to part a seal. Lay it on the opener rest along the desk, handle at the right end and blade pointing left.
8. Confirm the sheet tray, the card tray, and the recycling tray are empty and square to the desk, with the recycling tray at the front center, or at the front right for Config M.
9. Wipe the desk and clear away any paper scrap.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an

annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R2 (mail holder at the back right, recycling tray at the front center). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the desk — the mail holder is not in the start zone for the config, or the recycling tray is not where that config puts it; a gripper reaches across the desk for an envelope; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper takes each envelope to the work spot in Config M, the right gripper in Config R1 and R2; no arm reaches across the desk; the IF line followed is the one for the config on the desk).
- **Coaching note:** look where the mail holder is before the first reach, then follow that config's IF line through Step 1.

Violation: Envelopes taken out of order

- **Visible cue:** an envelope is taken from behind another one in the mail holder, or the holder is reordered before a pick.
- **SOP rule broken:** Step 1, the envelope at the front of the mail holder is the one taken.
- **Coaching note:** always take the front one. Never dig behind it.

Violation: A new envelope started before the last was finished

- **Visible cue:** a second envelope is brought to the work spot while the first sheet or card is still on the desk, or while the first empty envelope is still at the work spot.
- **SOP rule broken:** Steps 1 to 6, one envelope is run all the way through before the next one is taken.
- **Coaching note:** finish it. Card away, sheet away, envelope in the recycling tray, then take the next.

Violation: Envelope opened without the opener

- **Visible cue:** the flap is torn, ripped, peeled, or pulled open by a gripper, or a gripper works the corner gap open by hand instead of taking the opener.
- **SOP rule broken:** Step 2, the seal is parted by one draw of the opener under the flap.

- **Coaching note:** the opener does every opening. Nothing gets torn.

Violation: Envelope slit in the air

- **Visible cue:** the envelope is held up off the desk by one gripper while the other works the opener on it, or it is lifted clear part way through the draw.
- **SOP rule broken:** Step 2, the envelope lies flat at the work spot and is pinned while it is slit.
- **Coaching note:** it stays on the desk. Pin it and cut it there.

Violation: Envelope turned in the air

- **Visible cue:** the envelope is flipped, swung, or twisted over while held clear of the desk, instead of being lowered onto its bottom edge and swung flat there.
- **SOP rule broken:** Step 1, the envelope is laid down on the desk and tips over there.
- **Coaching note:** set the bottom edge down first, then walk the top edge over.

Violation: Slit walked, jabbed, or forced

- **Visible cue:** the opener is drawn in short tugs with releases in between, is lifted off the envelope part way along, is jabbed at the seal, or is forced on after stalling instead of being taken out and entered at the corner gap again.
- **SOP rule broken:** Step 2, one continuous draw from the corner gap to the left end with the tip under the flap the whole way.
- **Coaching note:** one run, all the way along, without letting go. If it stalls, come out and go in again.

Violation: Slit taken off the flap edge

- **Visible cue:** the tip goes into the envelope face or a short end rather than the corner gap, the draw wanders off the flap edge, or the sheet or the card inside comes out cut.
- **SOP rule broken:** Step 2, the tip enters the corner gap under the flap and runs along the flap edge.
- **Coaching note:** under the flap, along the top edge, nowhere else.

Violation: Opener not returned to its rest

- **Visible cue:** the opener is dropped on the desk, left lying at the work spot, still held while another thing is picked up, or laid on the rest turned the wrong way.
- **SOP rule broken:** Step 2, the opener goes back to tool home, handle at the right end and blade pointing left, before either gripper touches anything else.
- **Coaching note:** cut, then put it back the way you found it, then carry on.

Violation: Contents shaken or tipped out

- **Visible cue:** the envelope is lifted, tilted, shaken, or squeezed to empty it, or the mouth is not opened before the bundle is pulled at.

- **SOP rule broken:** Step 3, the cut flap is laid back and the bundle is drawn straight out of the mouth.
- **Coaching note:** open the mouth and pull the contents out. Never shake it.

Violation: Emptied or unfolded without a pin

- **Visible cue:** the left gripper is off the envelope while the bundle is drawn out, or off the base panel while a sheet panel is folded open, or it lifts away before the right gripper is clear, and the paper shifts under the work.
- **SOP rule broken:** Steps 3 and 5, the left gripper pins the envelope or the base panel and holds until the right gripper is off the paper.
- **Coaching note:** pin first, let go last.

Violation: Sheet unfolded before the card was taken off

- **Visible cue:** a sheet panel is folded open with the card still lying on the sheet, or the card is flicked or brushed aside instead of being lifted into the card tray.
- **SOP rule broken:** Steps 4 and 5, the card goes to the card tray before the sheet is unfolded.
- **Coaching note:** card away first, then open the sheet.

Violation: Sheet opened by dragging

- **Visible cue:** a panel is pushed, shoved, or slid across the desk to open it instead of being pinched, lifted clear, swung over its fold edge, and laid down.
- **SOP rule broken:** Step 5, each panel is lifted clear before it is swung over.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Sheet left folded or creased

- **Visible cue:** a sheet goes into the sheet tray with a panel still folded over, a corner curled, or a new crease across it; or a panel that sprang back is left as it is.
- **SOP rule broken:** Steps 5 and 6, all three panels lie flat in one line before the sheet is carried, and it goes into the tray fully open.
- **Coaching note:** flat and open before it leaves the work spot.

Violation: Wrong tray

- **Visible cue:** a sheet goes into the card tray, a card goes into the sheet tray, an envelope goes into either of them, or a sheet or card lands in the recycling tray.
- **SOP rule broken:** Steps 3, 4, and 6, sheets to the sheet tray, cards to the card tray, empty envelopes to the recycling tray.
- **Coaching note:** sheets back left, cards front left, envelopes front center.

Violation: Item stacked on another in a tray

- **Visible cue:** a sheet lands on top of a sheet already in the tray, or a card on top of a card, instead of clear of it.
- **SOP rule broken:** Steps 4 and 6, each item is laid clear of anything already in the tray.
- **Coaching note:** look at the tray before you lower. Put it in an empty part.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes an envelope from the mail holder, works the opener, slits, draws the contents out, unfolds the sheet, or presses the box; or the right gripper carries a sheet or a card to a tray. Both grippers taking a fold edge each in Step 9 is not this violation.
- **SOP rule broken:** Steps 1 to 9, the right gripper works and the left gripper pins and carries to the left trays.
- **Coaching note:** right works, left holds and takes the left trays.

Violation: Box worked before the envelopes were done

- **Visible cue:** a box flap is folded or the box is pressed while an envelope is still in the mail holder, or while a sheet or card is still at the work spot.
- **SOP rule broken:** Steps 7 to 9, the box is touched only after all three envelopes have been run.
- **Coaching note:** all three envelopes first. The box is last.

Violation: Box pressed before its top flaps were out

- **Visible cue:** the box is drawn flat with a top flap still standing up or folded inward into the box, so a flap creases or is trapped inside the flat box.
- **SOP rule broken:** Steps 7 and 8, all four top flaps lie flat out from their walls before the box comes down.
- **Coaching note:** all four flaps out first, then flatten.

Violation: Box shoved over or pressed from above

- **Visible cue:** the box is knocked, pushed, or swept over, is pressed straight down from above its top rim, or is worked flat in repeated shoves; or the near bottom crease is left unpinned and the box slides off the box spot.
- **SOP rule broken:** Step 8, the near bottom crease is pinned and the far top rim is drawn toward the front edge in one continuous run.
- **Coaching note:** hold the bottom and pull the top over. One steady run.

Violation: Fold edge left unstroked or a wall left standing

- **Visible cue:** the box is left with a wall standing up, a wall buckled, a bottom flap trapped folded under it, or a fold edge with no stroke drawn along it.

- **SOP rule broken:** Step 8, both fold edges are stroked end to end and every wall lies down flat.
- **Coaching note:** one run along each fold edge, and check that nothing is left standing.

Violation: Flat box carried or set down wrong

- **Visible cue:** the flat box is carried on one gripper, is tilted so a wall springs open, is released above the tray so it drops the rest of the way, lands skewed to the tray edges, overhangs a tray edge, or is left on the desk at the end of the episode.
- **SOP rule broken:** Step 9, both grippers take a fold edge each, carry the box level, lower it flat inside the tray, and release together.
- **Coaching note:** one hold each side, keep it level, all the way down, then let go together.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on an envelope, a sheet, a card, the opener, or the box without setting it down first.
- **SOP rule broken:** Steps 1 to 9, take a fresh hold only after the thing is resting.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens above the desk or a tray and the thing falls the rest of the way, or it opens over bare desk, over open space, or over the wrong zone.
- **SOP rule broken:** Steps 1 to 9, the gripper opens at the work spot, at tool home, at the box spot, or in a tray, once the thing is resting, and nowhere else.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, brushes, or straightens an envelope, a sheet, a card, or the box after releasing it, instead of lifting it clear and laying it down again.
- **SOP rule broken:** Steps 1 to 9, correct a bad placement by lifting the thing and laying it down again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Item dropped, collided, or knocked off the desk

- **Visible cue:** an envelope, a sheet, a card, the opener, or the box falls off the desk or out of a tray; the mail holder, the opener rest, or a tray is struck and shifts; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 9, carry on a clear path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an envelope still in the mail holder, a sheet or card still at the work spot, a sheet still folded in its tray, the opener off its rest, the box not flat or not in the recycling tray, or an arm away from home.
- **SOP rule broken:** Step 10, confirm the three trays, the opener at tool home, and the bare desk, then return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Envelope sealed shut at the corner gap**, so the opener tip cannot enter without being forced.
- **Envelope seal will not part** under two correct draws, or the paper tears along the draw rather than parting at the seal.
- **Defective opener:** the tip is bent, burred, or too blunt to part a seal under a correct draw.
- **Contents stuck inside the envelope**, glued to the seal or jammed, so the bundle will not draw out.
- **Sheet will not lie flat:** stroked along both fold edges and a panel still springs back every time.
- **Defective sheet or card:** found torn, damp, marked, split, or already creased off its fold lines.
- **Envelope loaded wrong:** it holds no sheet, no card, or an extra item, or the sheet is loaded with the wrong edge toward the flap.
- **Defective box:** crushed, taped, damp, or with a corner crease that will not fold, so it will not press flat or will not stand square.
- **Box or sheet is too large for its tray**, so it cannot lie inside without overhanging an edge.
- **Mail holder, opener rest, or a tray shifts out of place** during the episode.

Annotation subtasks (from SOP)

1. Take an envelope to the work spot
2. Slit the envelope with the opener and return the opener
3. Draw the contents out and clear the empty envelope
4. Put the card in the card tray

5. Unfold the sheet
6. Put the sheet in the sheet tray
7. Fold the box top flaps out
8. Press the box flat
9. Put the flat box in the recycling tray
10. Return both arms home and end the episode